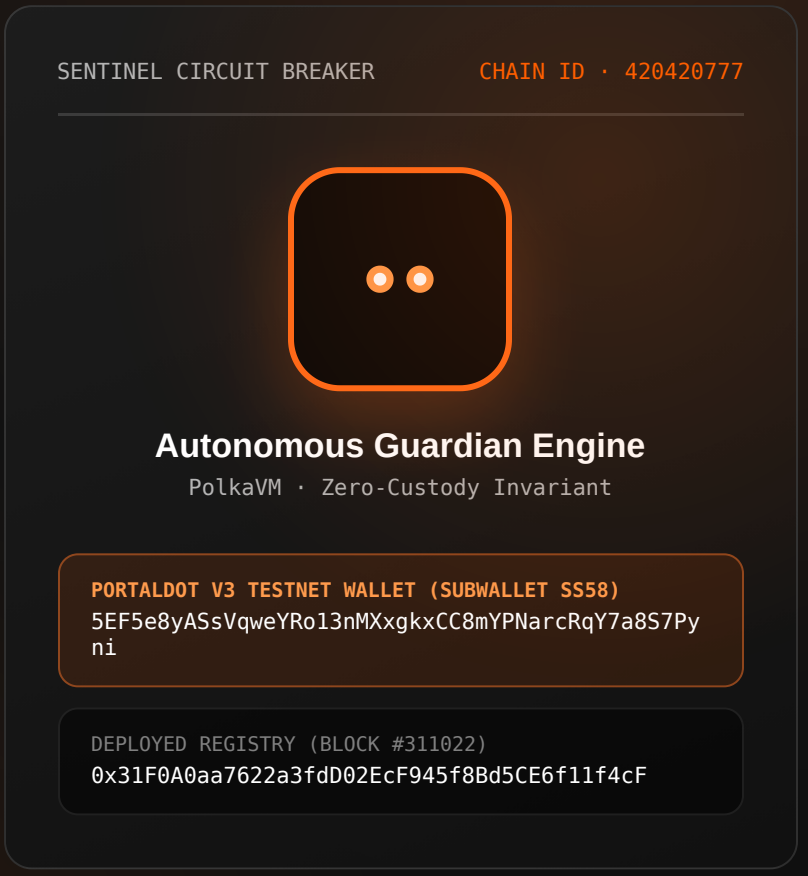
Transforming Block Speed into Defense

Track 3 · Tooling

4 Contracts

Live on Portaldot V3 Testnet

PortalSentinel is an autonomous closed-loop security firewall and block-level circuit breaker built natively for Portaldot V3 EVM (`pallet-revive` RISC-V PolkaVM), combining priority-fee defense with 24/7 Telegram mobile advisories.



Who We Are

Built for Modern On-Chain Security

A New Generation of Autonomous Node Defense

PortalSentinel is a closed-loop security daemon and smart-contract firewall engineered for Portaldot V3. We combine block-level anomaly detection, non-custodial Solidity circuit breakers, and operator CLI tooling to halt token approval drains automatically.

~1,000ms

Portaldot V3 `pallet-revive` Block Finality Cadence

12 / 12

Foundry Invariant & Unit Tests + 15 Daemon Tests Passing

1.35× – 1.85×

Priority Fee Escalation Multiplier for Emergency Preemption

DeFi users routinely grant unlimited ERC-20 allowances (`type(uint256).max`) to routers, bridges, and vaults. When a third-party contract is exploited, attackers drain wallets via `transferFrom` in seconds.

Gaps Between Detection and Execution

The Challenge

Human Latency Gap

Users are asleep or away from their desk. Manual revocation requires opening a browser, connecting a wallet, and signing—taking minutes while drains finish in seconds.

Oracle Consensus Lag (15–45 Seconds)

Multi-node oracle networks require Byzantine consensus rounds before firing a callback—arriving long after the attacker's drain transaction has finalized.

The Standard ERC-20 Revocation Flaw

Standard ERC-20 tokens only set allowance for `msg.sender`; an external guardian contract cannot overwrite a victim's allowance without a fallback capital sweep.

Silent Partial Revoke Failures

Naive security bots emit success events even when an underlying token reverts or ignores revocation, leaving victims exposed with a false sense of safety.

Passive alerts are post-mortems. Active sub-second intervention is essential.

End-to-End Closed-Loop Firewall

Our Approach

Block Telemetry & Scoring

Polls Portaldot V3 `pallet-revive` blocks with a crash-safe cursor journal, scoring exploit signals in <1ms against user sensitivity policies.

Verified Batch Revocation

Calls `guardianRevokeApproval` on revocable tokens and verifies post-call `allowance == 0` before counting success.

Emergency Capital Sweep

Evacuates standard ERC-20 balances via `SafeERC20` directly into the user's pre-registered cold vault when revoke is unsupported.

ACTION SELECTION ENGINE

On-Chain Pre-Check

Before broadcasting, the dispatcher inspects live `balanceOf` and `allowance` on Portaldot V3 so it never wastes gas on a no-op defense.

```
// Retry Escalation Ladder
Attempt 1 → 1.35× gas
Attempt 2 → 1.60× gas
Attempt 3 → 1.85× gas
```

Dual-Defense Architecture solves the standard ERC-20 revocation limitation.

Tailored Security Across Portaldot V3

What We Offer

PortalSentinel leverages Portaldot V3's dual Substrate/EVM runtime, storage-deposit economics, and native POT priority fee ordering.

pallet-revive RISC-V PolkaVM

Executes compiled smart contracts on RISC-V with ~1,000ms block cadence and deterministic gas—enabling rapid defensive settlement.

Storage-Deposit Optimized

Engineered for Portaldot V3's `DepositPerItem = 37.5 POT` model using compact Yul & Solidity bytecode to fit within faucet budgets.

SubWallet Testnet Integration

Native Portaldot V3 Testnet SubWallet connection & signature verification for account `5EF5e8yASsVqweYRo13nMXxgkxCC8mYPNarcRqY7a8S7Pyni` on Chain ID 420420777.

Zero-Custody Proof

Non-Custodial by Smart Contract Law

Even if the off-chain Guardian server is compromised, the `DefenseExecutor.sol` contract makes stealing user funds mathematically impossible: tokens can only move to the user's pre-registered `emergencySafe`.

```
// DefenseExecutor.sol — Enforced On-Chain Invariant
address safe = registry.getEmergencySafe(user);
if (safe == address(0) || safe == msg.sender) revert
InvalidEmergencySafe();
IERC20(token).safeTransferFrom(user, safe, balance);
```

0% Custody

Guardian Operator Can Never Receive Swept User Assets

ReentrancyGuard & Pausable

Built on OpenZeppelin v5 primitives with strict `onlyGuardian` and registry pause circuit breakers.

Verified Post-State Accounting

Reverts entire transaction (`NoDefenseExecuted`) if zero tokens were revoked or swept, preventing false-positive receipts.

Audited by 12 Foundry unit & invariant tests in contracts/test/PortalSentinel.t.sol

Built for Portaldot validator operators, security researchers, and everyday DeFi users—combining command-line node diagnostics, Prometheus telemetry, and 24/7 Telegram push alerts.

Node Tooling & 24/7 Mobile Autopilot

Track 3 Suite

- **Operator CLI Suite (`health · audit · benchmark · metrics`)**
  Probes Portaldot V3 EVM & Substrate RPC latency, audits wallet allowances on-chain, and benchmarks multi-sample p50/p95 block lag.
- **Prometheus Exporter (`/metrics` & `/health`)**
  Exposes real-time gauges and counters (`sentinel_monitored_block_height`, `sentinel_rpc_latency_ms`, `sentinel_defenses_executed_total`) for Grafana dashboards.
- **24/7 Telegram Bot Advisories (`@PortalSentinel_Alerts_bot`)**
  Pushes instant mobile alerts when a threat is detected and when an on-chain revoke or emergency sweep confirms, with direct explorer links.
- **Colocated Validator Docker Stack**
  Multi-stage `Dockerfile` and `docker-compose.yml` with persistent block-cursor volume (`/data/.sentinel-state.json`) and IPv4-first DNS tuning.

Live Telemetry Endpoints: portal-sentinel-pi.vercel.app/api/health & /api/metrics

Long-Term Roadmap Across Portaldot

P0 · Shipped on Testnet

All 4 contracts deployed on Portaldot V3 Testnet, post-call verified revocations, crash-safe block cursor, 27 automated tests, and live Vercel DApp.

P1 · General Calldata Detector

Decode arbitrary `transferFrom`, `permit`, and unverified spender interactions across all Portaldot DEXs & bridges with SQLite incident history.

P2 · Multi-Guardian Quorum

Threshold-signed guardian set colocated with Portaldot validators, per-user Telegram chat binding, and native integration into Portaldot Node Console.

Ecosystem Growth

Applying for the Portaldot Network Ecosystem Foundation Builder Program to scale from single-guardian testnet daemon to decentralized validator middleware.

Verified On-Chain

Live on Portaldot V3 Testnet

All four smart contracts are deployed and linked on Portaldot V3 Testnet (`Chain ID: 420420777`) via `pallet-revive`.

PORTALDOT V3 TESTNET WALLET ADDRESS (SUBWALLET SS58)
5EF5e8yASsVqweYRo13nMXxgkxCC8mYPNarcRqY7a8S7Pyni

LIVE WEB3 DAPP & DEMO VIDEO
https://portal-sentinel-pi.vercel.app ↗
https://portal-sentinel-pi.vercel.app/demo.mp4 ↗

PUBLIC GITHUB REPOSITORY
github.com/Blackwrld04/PortalSentinel ↗

SentinelRegistry.sol Block #311022
0x31F0A0aa7622a3fdD02EcF945f8Bd5CE6f11f4cF
Tx: 0xcf77198f0b1666c4084441d61db60a39f6a07981a9c9605fb4341689ba48336c

DefenseExecutor.sol Block #311067
0x89c9510c4b3AeEE8792De383D251ef0b61d406E2
Tx: 0x4cdd30974731623f7e9b2e25768d328669a4207ac5d0b7ea8aa0135ae31220f0

MockERC20.sol (pUSD) Block #311161
0x25da2DeD5b0Da07C160e021e0Df636988C3657f5
Tx: 0xf959f9ea03878c9ff0472ee3343f0c50c2be0a8fc119a41d6b83de2c250b515b

MockExploitFixture.sol Block #310782
0xc44a87FE907E497E50ffF6913c0b863bd0758a90
Tx: 0xcfdbdcf2d2807b19a5b8b00a5e4c0b7ee69cffd8e9df89be1c4e596a9fe1b941